A. Python IF (Single Condition)

1. Write a Python program to check if a number is positive.

```
a=int(input("Enter a positive number:"))
if a>0:
    print("number is positive",a)
print("number not positive")
```

2. Print "Eligible to vote" if age is 18 or above.

```
a=int(input("Enter the age  eligible for vote :"))
if a>18:
    print("Eligible for vote",a)
print("Not Eligible for vote")
```

3. Check if a number is divisible by 7.

```
a=int(input("Enter the number :"))
if a%7==0:
    print("divisible number",a)
print("Not divisible number")
```

4. Print "Pass" if marks are greater than 40

```
marks=int(input("Enter the marks :"))
if marks>=40:
    print("Pass")
print("Fail")
```

5. Check if a number is greater than 100.

```
num=int(input("Enter the number :"))
if num>100:
        print("This number is greater than 100")
print("This number is not greater than of 100")
```

6. Display a message if temperature exceeds 45°C.

```
temperature = int(input("Enter the temperature: "))
if temperature > 45:
    print("Temperature exceeds 45°C")
```

7. Check if a string length is more than 8 characters

```
 text =input("Enter a string: ")
if len(text) > 8:
    print("string length is more than 8 ")
```

8. Print "Logged In" if password matches "admin123".

```
password=input("Enter the password:")
if password=="admin123":
    print("Logged In")
```

9. Check if a number is a multiple of 10.

```
 num=int(input("Enter the number"))
if num%10==0:
    print("The number is a multiple of 10"
```

10. Print a warning if balance is below minimum limit.

```
balance=int(input("Enter a balance"))
minimum_limit=100
```

```
if balance<minimum_limit:
    print("Warning: Balance is below minimum limit")
```

B. Python IF–ELSE (Two Conditions)

11. Check whether a number is even or odd.

```
num = int(input("Enter a number: "))
if num % 2 == 0:
    print("The number is Even")
else:
    print("The number is Odd")
```

Output is:

Enter a number: 5

The number is odd

12. Find the largest of two numbers.

```
a = int(input("Enter a number: "))
b = int(input("Enter a number: "))
if a>b:
    print("a is largest number")
else:
    print("b is largest number")
```

Output is:

Enter a number: 9

Enter a number: 5

a is largest number

13. Check whether a person is eligible for driving license.

```
age = int(input("Enter a age: "))
if age>18:
    print("Eligible for driving lincense")
else:
    print("Not eligible for driving lincense")
```

Output is:

Enter a age: 13

Not eligible for driving lincense

14. Print "Pass" or "Fail" based on marks.

```
marks = int(input("Enter the marks: "))
if marks>40:
    print("Pass")
else:
    print("Fail")
```

Output is:

Enter the marks: 50

Pass

15. Check whether a number is positive or negative.

```
num = int(input("Enter the number: "))
if num>0:
    print("Number is positive")
else:
    print("Number is negative")
```

Output is:

Enter the number: -20

Number is negative

16. Check whether a character is a vowel or consonant.

```
ch = input("Enter a character: ")
if ch in ('a','e','i','o','u','A','E','I','O','U'):
    print("It is a Vowel")
else:
    print("It is a Consonant")
```

OutPut is :

Enter a character: f

It is a Consonant

17. Check if a year is leap or not.

18. Print "Valid Password" or "Invalid Password".

```
password=input("Enter the password:")
if password=="admin123":
    print("valid password")
```

```
else:
    print("invalid password")
```

Output is:

Enter the password:priyanka

invalid password

19. Determine whether salary is taxable or not.

```
salary = int(input("Enter your salary: "))
if salary > 500000:
    print("Salary is Taxable")
else:
    print("Salary is Not Taxable")
```

Output is:

Enter your salary: 100000

Salary is Not Taxable

20. Check whether a number is greater than 50 or not

```
num = int(input("Enter the number: "))
if num>50:
    print("Number is greater of 50")
else:
    print("Number is not greater than of 50")
```

Output is :

Enter the number: 40

Number is not greater than of 50

C. Python NESTED IF–ELSE

21. Find the largest of three numbers.

```
a=int(input("Enter a number:"))
b=int(input("Enter a number:"))
c=int(input("Enter a number:"))
if a>b:
    if a>c:
        print("a is greater")
    else:
        print("c is greater")
else:
    if b>c:
        print("b is the greater")
    else:
        print("c is the greater")
```

22. Check whether a number is positive, negative, or zero.

```
num = float(input("Enter a number: "))


if num >= 0:
    if num == 0:
```

```
        print("The number is Zero")
    else:
        print("The number is Positive")
else:
    print("The number is Negative")
```

Output is:

Enter a number: 0

The number is Zero

23. Assign grades:

- **A → marks ≥ 90**
- **B → marks ≥ 75**
- **C → marks ≥ 60**
- **Fail → below 60**

```
marks = int(input("Enter your marks: "))
if marks >= 90:
    print("Grade A")
else:
    if marks >= 75:
        print("Grade B")
    else:
        if marks >= 60:
            print("Grade C")
```

```
    else:
        print("Fail")
```

Output:

Enter your marks: 50

Fail

24. Check whether a triangle is equilateral, isosceles, or scalene.

```
a = int(input("Enter first side: "))
b = int(input("Enter second side: "))
c = int(input("Enter third side: "))
if a == b:
    if b == c:
        print("Triangle is Equilateral")
    else:
        print("Triangle is Isosceles")
else:
    if a == c or b == c:
        print("Triangle is Isosceles")
    else:
        print("Triangle is Scalene")
```

Output is:

Enter first side: 4

Enter second side: 5

Enter third side: 4

Triangle is Isosceles

25. Check whether a character is uppercase, lowercase, digit, or special character.

```
ch = (input("Enter the character "))
if ch.isalpha():
    if ch.isupper():
        print("UpperCase Letter")
    else:
        print("LowerCase Letter")
else:
    if ch.isdigit():
        print("Digit")
    else:
        print("Special character")
```

Output is:

Enter the character 123

Digit

Second Way

```
ch=input("Enter the character:")
if ch>='0':
    if ch<='9':
        print("digit")
    else:
```

```
        if ch<'Z':
            print("Upper case")
        else:
            if ch<'z':
                print("Lower case")
            else:
                print("Special character")
else:
    print("Special character")
```

Another way solution through the complex:

```
ch=input("Enter the character:")
if ch>='0' and ch<='9':
    print("Digit")
elif ch>='A' and ch<='Z':
    print("Upper case")
elif ch>='a' and ch<='z':
    print("Lower case")
else:
    print("Special character")
```

26. Calculate electricity bill using slab-wise rates.

```
#service charge=250 ,1-1-100=>5%,101-200=>7/-,200 +=>10/-
service =200
```

```
unit=int(input("Enter the unit:"))
if unit<=100:
   bill=unit*5
else:
   if unit<=200:
      bill*7
   else:
      bill=unit*10
bill=bill+service
print(bill)
```

Output is:

```
Enter the unit:5
225
```

27. Validate login using username and password.

```
username = input("Enter username: ")
password = input("Enter password: ")
if username == "priyanka":
   if password == "soni123":
      print("Login Successful")
   else:
      print("Incorrect Password")
else:
   print("Username and Password Wrong")
```

Output is:

Enter username: priyanka

Enter password: soni123

Login Successful

28. Check student result using marks of 3 subjects.

```
physics_marks = int(input("Enter your physics marks: "))
chemistry_marks = int(input("Enter your chemistry marks: "))
math_marks = int(input("Enter your math marks: "))
if physics_marks >= 33:
    print("pass")
    if chemistry_marks >=33:
        print("pass")
    else:
        if math_marks>=33:
            print("pass")
        else:
            print("Fail")
else:
    print("Fail")
```

second way *******Three condition are true***

```
physics_marks = int(input("Enter your physics marks: "))
chemistry_marks = int(input("Enter your chemistry marks: "))
```

```
math_marks = int(input("Enter your math marks: "))
if physics_marks >= 33:
    if chemistry_marks >=33:
        if math_marks>=33:
            print("pass")
        else:
            print("Fail")
else:
    print("Fail")
```

Output is:

```
 Enter your physics marks: 33
Enter your chemistry marks: 33
Enter your math marks: 20
Fail
```

#29. Find the second largest number among three numbers.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))
if a > b:
    if b > c:
        print("Second largest number is:", b)
```

```
    else:
        if a > c:
            print("Second largest number is:", c)
        else:
            print("Second largest number is:", a)
else:
    if a > c:
        print("Second largest number is:", a)
    else:
        if b > c:
            print("Second largest number is:", c)
        else:
            print("Second largest number is:", b)
```

30. Check loan eligibility using age, salary, and credit score

```
age = int(input("Enter your age: "))
salary = int(input("Enter your salary: "))
credit_score = int(input("Enter your credit score: "))
if age >= 25:
    if salary >= 30000:
        if credit_score >= 800:
            print("Loan Approved")
```

```
        else:
            print("Loan Not Approved")
    else:
        print("Loan Not Approved")
else:
    print("Loan Not Approved")
```

Output is:

```
Enter your age: 30
Enter your salary: 35000
Enter your credit score: 500
Loan Not Approved
```

D. Python ELIF (Multiple Conditions)

31. Print day name using day number (1–7).

```
day=int(input("Enter the day number:"))
if day==1:
    print("Monday")
elif day==2:
    print("Tuesday")
elif day==3:
    print("Wednesday")
```

```
elif day==4:
    print("Thursday")
elif day==5:
    print("Friday")
elif day==6:
    print("Saturday")
elif day==7:
    print("Sunday")
else:
    print("Invalid day number")
```

Output is:

```
Enter the day number:5
Friday
```

32. Print month name using month number.

```
month = int(input("Enter month number:"))
if month == 1:
    print("January")
elif month == 2:
    print("February")
elif month == 3:
    print("March")
```

```
elif month == 4:
    print("April")
elif month == 5:
    print("May")
elif month == 6:
    print("June")
elif month == 7:
    print("July")
elif month == 8:
    print("August")
elif month == 9:
    print("September")
elif month == 10:
    print("October")
elif month == 11:
    print("November")
elif month == 12:
    print("December")
else:
    print("Invalid month number")
```

Output is:

Enter month number:9

September

33. Display grade based on percentage.

```
per = float(input("Enter the percentage: "))
if per >= 90:
   print("Grade A")
elif per >= 75:
   print("Grade B")
elif per >= 60:
   print("Grade C")
elif per >= 50:
   print("Grade D")
else:
   print("Fail")
```

Output is:

Enter your percentage: 76

Grade B

34. Display bonus percentage based on experience years.

```
exp = float(input("Enter the experience: ")
if exp >12:
   print("bonus is :20%")
```

```
elif exp >5:
    print("bonus is :7%")
elif exp >2:
    print("bonus is :2%")
else:
    print("No bonus")
```

Output is:

Enter the experience: 5

bonus is :2%

35. Identify traffic signal meaning.

```
signal=input("Enter the traffic signal identity:")
if signal=="green":
    print("Go")
elif signal=="red":
    print("Stop")
elif signal=="yellow":
    print("Wait")
else:
    print("signal not valid")
```

Output is:

Enter the traffic signal identity:red

Stop

36. Categorize temperature as Cold / Warm / Hot.

```
temp=int(input("Enter the tempature:"))
if temp<=15:
   print("Cold")
elif temp<=30:
   print("Warm")
else:
   print("Hot")
```

Output is:

Enter the tempature:31

Hot

37. Categorize employee based on salary range.

```
        salary=int(input("Enter the employee salary:"))
if salary>=80000:
   print("High salary employee")
elif salary >=50000:
   print("Medium level salary employee")
elif salary>=20000:
   print("low level salary employee")
else:
```

```
    print("very low level salary")
```

Output is:

Enter the employee salary:60000

Medium level salary employee

38. Print discount percentage based on purchase amount.

```
amount = int(input("Enter the purchase amount: "))
if amount >=12000:
    print("Discount is :20%")
elif amount >=5000:
    print("Discount is :10%")
elif amount >=2000:
    print("Discount is :2%")
else:
    print("No Discount")
```

Output is:

Enter the purchase amount: 7000

Discount is :10%

39. Identify number type: single-digit / double-digit / multi-digit.

```
num = int(input("Enter a number: "))
```

```
if num >= -9 and num <= 9:
    print("Single digit number")
elif (num >= 10 and num <= 99) or (num <= -10 and num >= -99):
    print("Double digit number")
else:
    print("Multi digit number")
```

Output is:

Enter a number: 11

Double-digit number

40. Assign performance rating: Poor / Average / Good / Excellent

```
rating = int(input("Enter the perfotmance rating: "))
if rating >=90:
    print("Excellent")
elif rating >=75:
    print("Good")
elif rating >=50:
    print("Average")
else:
    print("poor")
```

Output is:

Enter the perfotmance rating: 40

Poor

E. Python COMPLEX CONDITIONS (AND / OR / NOT)

41. Check whether a number is divisible by 5 and 11.

```
num = int(input("Enter a number: "))
if num % 5 == 0 and num % 11 == 0:
    print("Number is divisible by 5 and 11")
else:
    print("Number is not divisible by 5 and 11")
```

Output is:

Enter a number: 55

Number is divisible by 5 and 11

42. Check if a person is eligible for loan: ● age ≥ 21 ● salary ≥ 25,000 ● credit score ≥ 700

```
age = int(input("Enter a age: "))
salary = int(input("Enter a salary: "))
creditScore = int(input("Enter the creditScore: "))
if age>=21 and salary>=2500 and creditScore>=700:
    print("person is eligible for loan")
```

```
else:
    print("person is not eligible for loan")
```

Output is:

Enter a age: 21

Enter a salary: 25000

Enter the creditScore: 800

person is eligible for loan

43. Validate login using username AND password.

```
username =input("Enter a username: ")
password =input("Enter a password: ")
if username=="priyanka" and password=="deep123":
    print("Login Successful")
else:
    print("Invalid Username or Password")
```

Output is:

Enter a username: priyanka

Enter a password: deep123

Login Successful

44. Check student pass condition: ● All subjects ≥ 40 ● Average ≥ 50

```
m1 = int(input("Enter marks of subject 1: "))
```

```
m2 = int(input("Enter marks of subject 2: "))
m3 = int(input("Enter marks of subject 3: "))
avg = (m1 + m2 + m3) / 3
if m1 >= 40 and m2 >= 40 and m3 >= 40 and avg >= 50:
    print("Student Passed")
else:
    print("Student Failed")
```

Output is:

```
Enter marks of subject 1: 51
Enter marks of subject 2: 43
Enter marks of subject 3: 44
Student Failed
```

45. Check if a number lies between 10 and 100.

```
number=int(input("Enter the number:"))
if number>=10 and number<=100:
    print("number is between 1 to 100")
else:
    print("number is not between 1 to 100")
```

Output is:

```
Enter the number:101
number is not between 1 to 100
```

46. Check exam eligibility: ● attendance ≥ 75% OR ● medical certificate available

```
attendance = int(input("Enter the attendance percentage: "))
medical = input("Medical certificate available(yes/no):")
if attendance >= 75 or medical == "yes":
    print("Student is eligible for exam")
else:
    print("Student is not eligible for exam")
```

Output is:

```
Enter the attendance percentage: 50
Medical certificate available(yes/no):yes
Student is eligible for exam
```

47. Validate a date using conditions.

```
day = int(input("Enter day: "))
month = int(input("Enter month: "))
year = int(input("Enter year: "))
if day >= 1 and day <= 31 and month >= 1 and month <= 12 and year > 0:
    print("Valid Date")
else:
    print("Invalid Date")
```

Output is:

Enter day: 3

Enter month: 5

Enter year: 2024

Valid Date

48. Check whether an email format is valid.

```
email = input("Enter the email: ")
if "@" in email and "." in email:
    print("Valid Email")
else:
    print("Invalid Email")
```

Output is:

Enter your email:

priyanka.kamboj@gmail.com

Valid Email

49. Determine insurance eligibility using age, health status, and income.

```
age = int(input("Enter the age:"))
health = input("Enter health status (good/bad): ")
income = int(input("Enter the income:"))
if age >= 18 and health == "good" and income >= 20000:
```

```
    print("Eligible for Insurance")
else:
    print("Not Eligible for Insurance")
```

Output is:

Enter the age:10

Enter health status (good/bad): bad

Enter the income:90000

Not Eligible for Insurance

50. Check leap year using complete leap year logic.

```
year = int(input("Enter a year: "))
if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Leap Year")
else:
    print("Not a Leap Year")
```

Output is:

Enter a year: 2024

Leap Year

F. INTERVIEW-LEVEL PYTHON LOGIC QUESTIONS

51. Write a Python program to calculate income tax using slabs.

```
income = float(input("Enter your annual income: "))
if income <= 250000:
    tax = 0
elif income <= 500000:
    tax = income * 0.05
elif income <= 1000000:
    tax = income * 0.20
else:
    tax = income * 0.30
print("Income Tax =", tax)
```

52. Create an ATM withdrawal program with balance checks.

```
bal=30000
print("1.check balance\2.Withrawal balance\3.Exit")
ch=int(input("Enter your choice:"))
if ch==1:
    print("Your current balance:",bal)
elif ch==2:
    w=int(input("Enter amount to withrawal:"))
    if w<=bal:
        print("Balance withrawal is:",w)
        print("Remaning amount:",(bal-w))
```

```
    else:
        print("Insuffient balance")
else:
    print("Your choice is wrong")
```

Output is:

1.check balance.Withrawal balance——————————.Exit

Enter your choice:2

Enter amount to withrawal:2500

Balance withrawal 2500

Remaning amount: 27500

53. Check promotion eligibility using experience and performance.

```
experience = int(input("Enter years of experience: "))
performance = int(input("Enter performance score: "))
if experience >= 5 and performance >= 80:
    print("Eligible for Promotion")
else:
    print("Not Eligible for Promotion")
```

Output is:

Enter years of experience: 6

Enter performance score: 90

Eligible for Promotion

54. Implement a grading system using nested if–else.

```
marks = int(input("Enter your marks: "))
if marks >= 90:
    print("Grade A")
else:
    if marks >= 75:
        print("Grade B")
    else:
        if marks >= 60:
            print("Grade C")
        else:
            print("Fail")
```

Output is:

Enter your marks: 65

Grade C

55. Validate strong password using multiple conditions.

```
password = input("Enter your password: ")


lower = False
upper = False
```

```
digit = False
special = False

for ch in password:
    if ch >= 'a' and ch <= 'z':
        lower = True
    elif ch >= 'A' and ch <= 'Z':
        upper = True
    elif ch >= '0' and ch <= '9':
        digit = True
    elif (ch == '!' or ch == '@' or ch == '#' or ch == '$' or ch == '%' or ch == '^' or ch == '&' or ch == '*'):
        special = True
if (len(password) >= 8 and lower and upper and digit and special):
    print("Strong Password ")
else:
    print("Weak Password")
```

Output is:

Enter your password: Priyanka2@123

Strong Password

56. Calculate delivery charges based on location and order amount.

```
location = input("Enter location (local/outside): ")
amount = float(input("Enter order amount: "))
if location == "local" and amount >= 500:
    charge = 0
elif location == "local" and amount < 500:
    charge = 50
elif location == "outside" and amount >= 1000:
    charge = 50
else:
    charge = 100
print("Delivery Charge =", charge)
```

Output is:

```
Enter location (local/outside): local
Enter order amount: 15000
Delivery Charge = 0
```

57. Determine online exam qualification.

```
attendance = int(input("Enter attendance percentage: "))
assignment = input("Assignment submitted (yes/no): ")
if attendance >= 75 and assignment == "yes":
```

```
    print("Eligible for Online Exam")
else:
    print("Not Eligible for Online Exam")
```

Output is:

Enter attendance percentage: 90

Assignment submitted (yes/no): yes

Eligible for Online Exam

58. Create movie ticket pricing logic based on age & show time.

```
age = int(input("Enter age: "))
show = input("Enter show time (day/night): ")
if age < 12:
    price = 100
else:
    if age < 60:
        price = 200
    else:
        price = 150
if show == "night":
    price = price + 50
print("Ticket Price =", price)
```

Output is:

Enter age: 60

Enter show time (day/night): day

Ticket Price = 150

59. Determine bank account type based on balance.

```
balance = float(input("Enter your account balance: "))
if balance >= 100000:
    print("Premium Account")
elif balance >= 50000:
    print("Gold Account")
elif balance >= 10000:
    print("Savings Account")
else:
    print("Basic Account")
```

Output is:

Enter your account balance: 500000

Premium Account

60. Create a menu-driven program using if–elif–else

```
print("Menu")
print("1. Addition")
print("2. Subtraction")
```

```
print("3. Multiplication")
print("4. Division")
choice = int(input("Enter your choice (1-4): "))
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
if choice == 1:
    print("Addition =", a + b)
elif choice == 2:
    print("Subtraction =", a - b)
elif choice == 3:
    print("Multiplication =", a * b)
elif choice == 4:
    print("Division =", a / b)
else:
    print("Invalid Choice")
```

Output is:

```
Menu
1. Addition
2. Subtraction
3. Multiplication
4. Division
```

Enter your choice (1-4): 3

Enter first number: 5

Enter second number: 3

Multiplication = 15